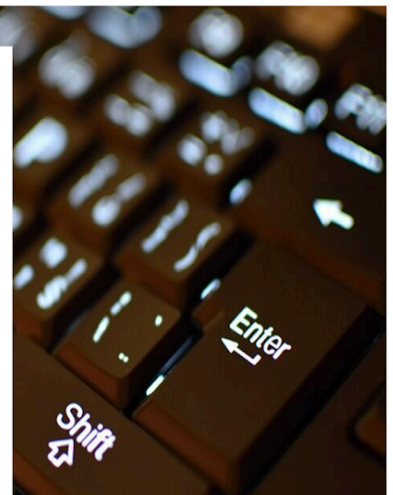


Computer Awareness

STUDY NOTES

Computer Hardware



Computer Hardware

Hardware

Hardware refers to the physical and tangible parts of a computer system, i.e. the components that can be seen and touched by the user. Hardware can be taken as the body of a computer consisting of different logically and physically operational parts for different functions.

Following can be examples of Hardware:

- Input devices – keyboard, mouse, etc.
- Output devices – printer, monitor, etc.
- Secondary storage devices – Hard disk, CD, DVD, etc.
- Internal components – CPU, motherboard, RAM, etc.

Relationship between Hardware and Software

- Both Hardware and Software are mutually dependent on each other. Both of them should work in collaboration in order to make a computer produce a useful output.
- Software cannot be used without equivalent hardware.
- Hardware in absence of a set of programs to operate upon cannot perform any useful operations and is useless.
- To get a particular task or operation executed by the computer, relevant software must be loaded into the hardware.
- Hardware is a one-time investment.
- Development of Software is comparatively expensive and is a prolonged expense.
- Various software applications can be installed on hardware to perform various tasks.
- A software works as an interface between the hardware and its user.
- If the hardware can be considered as the 'heart' of a computer system, then the software is considered as its 'soul'. Both of them are complementary to each other.

CPU

Central processing unit (**CPU**), principal part of any digital computer system, generally composed of the main memory, control unit, and arithmetic-logic unit.

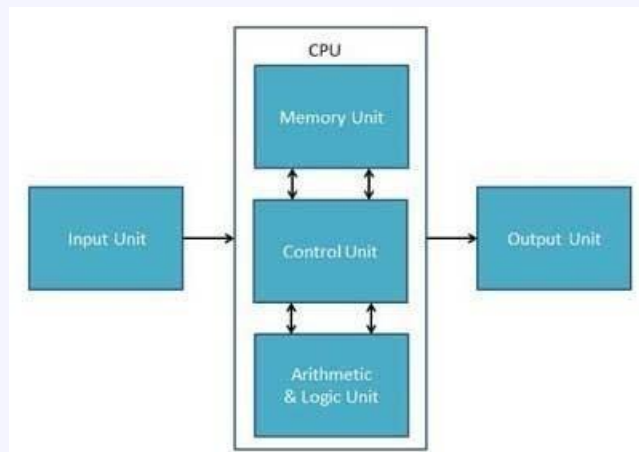
- Central Processing Unit (CPU) consists of the following features –
- CPU is considered as the brain of the computer.

- CPU performs all types of data processing operations.
- It stores data, intermediate results, and instructions (program).
- It controls the operation of all parts of the computer.



The CPU itself has the following three components.

- Memory or Storage Unit
- Control Unit
- ALU(Arithmetic Logic Unit)



Memory or Storage Unit

This unit can store instructions, data, and intermediate results. This unit supplies information to other units of the computer when needed. It is also known as the internal storage unit or the main memory or the primary storage or Random Access Memory (RAM).

Its size affects speed, power, and capability. Primary memory and secondary memory are two types of memories in the computer. Functions of the memory unit are –

- It stores all the data and the instructions required for processing.
- It stores intermediate results of processing.

- It stores the final results of processing before these results are released to an output device.
- All inputs and outputs are transmitted through the main memory.

Control Unit

This unit controls the operations of all parts of the computer but does not carry out any actual data processing operations.

Functions of this unit are –

- It is responsible for controlling the transfer of data and instructions among other units of a computer.
- It manages and coordinates all the units of the computer.
- It obtains the instructions from the memory, interprets them, and directs the operation of the computer.
- It communicates with Input/Output devices for transfer of data or results from storage.
- It does not process or store data.

ALU (Arithmetic Logic Unit)

This unit consists of two subsections namely,

- Arithmetic Section
- Logic Section

Arithmetic Section

The function of the arithmetic section is to perform arithmetic operations like addition, subtraction, multiplication, and division. All complex operations are done by making repetitive use of the above operations.

Logic Section

The function of the logic section is to perform logic operations such as comparing, selecting, matching, and merging of data.

What is the latest computer processor?

- Best CPU: AMD Ryzen 9 5900X. ...
- Best high-end CPU: AMD Ryzen 9 3950X. ...
- Best mid-range CPU: AMD Ryzen 5 3600X. ...
- Best entry-level CPU: AMD Ryzen 3 3100. ...
- Best gaming CPU: AMD Ryzen 7 5800X. ...
- Best VR CPU: Intel Core i5-10600K. ...
- Best performance **processor**: AMD Ryzen Threadripper 3960X. ...
- Best budget CPU: AMD Athlon 300GE.

Motherboard

- The motherboard works as a single platform to assemble all of the parts of a computer system together. It connects the primary components like CPU, memory, hard drives, and the secondary components like optical drives, video card, sound card, and other ports and expansion cards directly or via cables. It acts as the backbone of a computer.



Features of Motherboard

A motherboard comes with various features as follows:

- Motherboards vary very much in supporting different types of devices and components.
- Motherboard provides provision for a single type of CPU and a multiple number of memories.
- Devices like video cards, hard disks, sound cards have to be compatible with the particular motherboard to integrate with the rest of the components.
- Motherboards, cases, and power supplies should be compatible to work properly in collaboration with the other components.

Popular Manufactures

Following are few of the popular manufacturers in the market of the motherboard:

- Intel
- ASUS
- A Open
- ABIT
- Bio star
- Gigabyte
- MSI

Description of Motherboard

- The motherboard is installed inside the case or cabinet and is securely attached using small screws. The motherboard consists of ports to connect all of the internal hardware components. It has provision for a single socket for CPU, whereas for memory, generally one or more slots are available. Motherboards contain ports to attach input devices like the floppy drive, hard drive, and optical drives via ribbon cables. The motherboard carries fans for heat dissipation and a special component called SMPS (switched-mode power supply) designed for power supply.
- It contains a peripheral card slot in front of the motherboard which can be used to attach output/input devices like video cards, sound cards, and other expansion cards to the motherboard.
- On the other side, motherboards carry a variety of ports to connect the important input/output devices like monitor, printer, mouse, keyboard, speaker, and network cables. Motherboards also consist of advanced technology known as USB ports, which allow compatible devices to be connected in plug-in/plug-out fashion. For example, pen drive, digital cameras, etc.

RAM

RAM, which stands for **random access memory**, and **ROM**, which stands for read-only memory, are both present in your computer. **RAM** is volatile memory that temporarily stores the files you are working on. **ROM** is non-volatile memory that permanently stores instructions for your computer. The information stored in RAM can be checked with the help of BIOS. It is generally known as the main memory or temporary memory or cache memory or volatile memory of the computer system.

Two main types of RAM are:

- Static RAM
- Dynamic RAM

Static RAM

Static RAM is the full form of SRAM. In this type of RAM, data is stored using the state of a six transistor memory cell. Static RAM is mostly used as a cache memory for the processor (CPU).

Dynamic RAM

- DRAM stands for Dynamic Random Access Memory. It is a type of RAM which allows you to store each bit of data in a separate capacitor within a

specific integrated circuit. Dynamic RAM is a standard computer memory of the many modern desktop computers.

- This type of RAM is a volatile memory that needs to be refreshed with voltage regularly. Else it loses the information stored on it.

SRAM VS DRAM

SRAM	DRAM
SRAM is costlier than DRAM.	DRAM costs less compared to SRAM.
It is a complex internal circuitry, and it offers less storage capacity than is available compared to the same physical size of a DRAM memory chip.	It is the small internal circuitry in the one-bit memory cell of DRAM. The large storage capacity is available.

Types of RAM

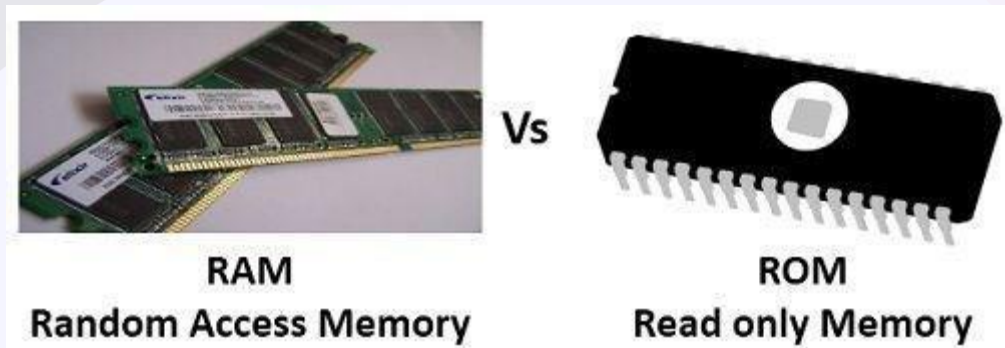
Type of RAM	Year Invented
EDO RAM (Extended data operations read-only memory)	1994
RDRAM (Rambus RAM)	1998
DDR2	2003
DDR4	2012

ROM

- Read-Only **Memory** (ROM), is a type of electronic storage that comes built into a device during manufacturing. You'll find ROM chips in computers and many other types of electronic products; VCRs, game

consoles, and car radios all use ROM to complete their functions smoothly.

- ROM chips come built into an external unit – like flash drives and other auxiliary memory devices or installed into the device's hardware on a removable chip. Non-volatile memory like ROM remains viable even without a power supply.



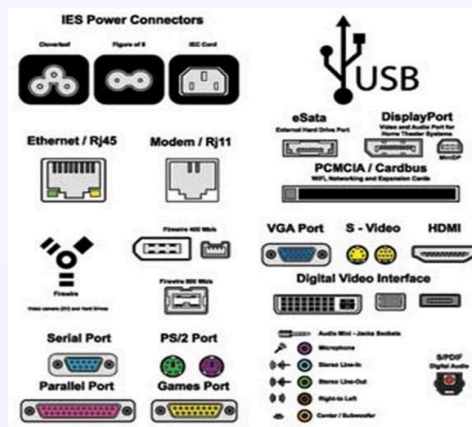
Ports

- In terms of computers, a port is a physical docking socket using which an external or internal device can be connected to the motherboard. It may also be a programmatic docking point which can be used for flow of information from a program to the computer or over the Internet.

Characteristics of Ports

Following are the characteristics of a port:

- Most of the external devices are connected to the system through cables and ports.
- Ports are sockets on the motherboard of the system into which the device itself or the cable of the external device is plugged in.
- Examples: mouse, keyboard, monitor, microphone, speakers, etc.



Serial Port

- Useful for external modems and older version of computer mouse
- Two possible versions: 9 pin, 25 pin model
- Data travels at the speed of 115 kilobits per second

Parallel Port

- Used to attach scanners and printers
- Also known as printer port
- 25 pin model
- IEEE 1284-compliant Centronics port

PS/2 Port

- Used to connect earlier version computer keyboard and mouse
- Also known as mouse port
- Most of the earlier version computers provide two PS/2 port, each for the mouse and keyboard
- IEEE 1284-compliant Centronics port

Universal Serial Bus (USB) Port

- It can be used to attach a variety of external USB devices like external hard disk, printer, scanner, mouse, keyboard, etc.
- It was introduced in 1997.
- Most of the newer versions of computers are equipped with two USB ports as minimum.
- Data travels at the speed of 12 megabits per seconds.
- USB compliant devices can receive power from a USB port.

VGA Port

- It is used to connect a monitor to a computer's video card.
- It consists of 15 holes.
- It resembles the serial port connector. However, the serial port connector consists of pins, and the VGA port consists of holes.

Power Connector

- Three-pronged plug.
- It connects to the computer's power socket available in SMPS that plugs into a power bar or wall socket.

FireWire Port

- It can transfer huge amounts of data at comparatively very fast speed.
- It can connect camcorders and video equipment to the computer system.
- Data flows at the speed of 400 to 800 megabits per seconds.
- Invented by Apple.

- It has 3 variants: 4-Pin FireWire 400 connector, 6-Pin FireWire 400 connector, and 9-Pin FireWire 800 connector.

Modem Port

- It is used to connect a PC's modem to the telephone network.

Ethernet Port

- Used to connect computer systems to a network and high speed Internet.
- Connects the network cable to the system.
- This port is available on the Ethernet Card.
- Data travels at the speed of 10 megabits to 1000 megabits per seconds, which depends upon the network bandwidth.

Game Port

- Used to connect a joystick to a PC
- It is now replaced by USB

Digital Video Interface, DVI Port

- Used to connect digital signal accepting Flat panel LCD monitors to the computer's high-end video graphic cards.
- Very popular nowadays among video card manufacturers.

Sockets

- Sockets usually connect the microphone (mic) and speakers to the sound card on the motherboard.